

3 Fundamental Data Structures

Efficient Algorithms · Wintersemester 2026–27

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Learning Outcomes

Unit 3: ***Fundamental Data Structures***

1. Understand and demonstrate the difference between *abstract data type (ADT)* and its *implementation*
2. Be able to define the ADTs *stack*, *queue*, *priority queue* and *dictionary/symbol table*
3. Understand *array*-based implementations of stack and queue
4. Understand *linked lists* and the corresponding implementations of stack and queue
5. Know *binary heaps* and their performance characteristics
6. Understand *binary search trees* and their performance characteristics
7. Know high-level idea of basic *hashing strategies* and their performance characteristics

Outline

3 Fundamental Data Structures

3.1 Stacks & Queues
3.2 Resizable Arrays
3.3 Priority Queues & Binary Heaps
3.4 Operations on Binary Heaps
3.5 Symbol Tables
3.6 Binary Search Trees
3.7 Ordered Symbol Tables
3.8 Balanced BSTs
3.9 Hashing – Basics
3.10 Universal Hashing

Recap: The Random Access Machine

- Data structures make heavy use of pointers and dynamically allocated memory.

- Recall: Our RAM model supports
  - basic pseudocode (≈ simple Python/Java code)
  - creating arrays of a fixed/known size.
  - creating instances (objects) of a known class.



Python abstracts this away!

no predefined capacity!

There are *no arrays* in Python, only its built-in *lists*.

But: Python *implementations create* lists based on fixed-size arrays (stay tuned!)



Python ≠ RAM: Not every built-in Python instruction runs in $O(1)$ time!

3.1 Stacks & Queues

Abstract Data Types

abstract data type (ADT)

- list of supported operations
- **what** should happen
- **not:** how to do it
- **not:** how to store data

≈ Java `interface`, Python `ABC`s (with comments)

abstract base classes

VS.

data structures

- specify exactly **how** data is represented
- **algorithms** for operations
- has concrete costs (space and running time)

≈ Java/Python `class` (non abstract)

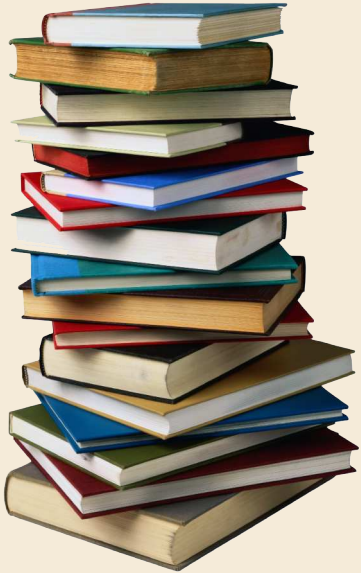
Why separate?

- Can swap out implementations ⇝ "drop-in replacements"

⇝ **reusable code**!

- (Often) better abstractions
- Prove generic lower bounds (⇝ Unit 4)

Stacks



Stack ADT

- `top()`
  Return the topmost item on the stack
  Does not modify the stack.
- `push(`x`)`
  Add x onto the top of the stack.
- `pop()`
  Remove the topmost item from the stack
  (and return it).
- `isEmpty()`
  Returns `true` iff stack is empty.
- `create()`
  Create and return an new empty stack.

Linked-list implementation for Stack

Invariants:

- maintain pointer *top* to topmost element
- each element points to the element below it (or `null` if bottommost)

```
1  class Node
2      value
3      next
4
5  class Stack
6      top := null
7      procedure top():
8          return top.value
9      procedure push(x):
10         top := new Node(x, top)
11     procedure pop():
12         t := top()
13         top := top.next
14         return t
```

Linked-list implementation for Stack – Discussion

Linked stacks:


require $\Theta(n)$ space when n elements on stack


All operations take $O(1)$ time


require $\Theta(n)$ space when n elements on stack

Can we avoid extra space for pointers?

Array-based implementation for Stack

If we want no pointers $\rightsquigarrow$ array-based implementation

Invariants:

- maintain array S of elements, from bottommost to topmost
- maintain index *top* of position of topmost element in `S`.



What to do if stack is full upon `push`?

Array stacks:

- require *fixed capacity* C (decided at creation time)!
- require $\Theta(C)$ space for a capacity of C elements
- all operations take $O(1)$ time

Queues

Operations:

- `enqueue(`x`)`
  Add x at the end of the queue.
- `dequeue()`
  Remove item at the front of the queue and return it.



Implementations similar to stacks.

Bags

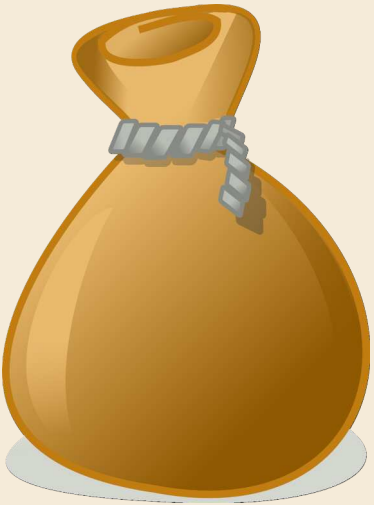
What do Stack and Queue have in common?

They are special cases of a ***Bag***!

Update Operations:

- `insert(`x`)`
  Add x to the items in the bag.
- `delAny()`
  Remove any one item from the bag and return it.
  (Not specified which; any choice is fine.)

- roughly similar to Java's `java.util.Collection`
  Python's `collections.abc.Collection`
- always support iterating over content (read only)



Sometimes it is useful to *state* that order is irrelevant $\rightsquigarrow$ Bag
Implementation of Bag usually just a Stack or a Queue

3.2 Resizable Arrays

Digression – Arrays as ADT

Arrays can also be seen as an ADT! . . . but are commonly seen as specific data structure

Array operations:

- `create(`n`)` *Java*: `A = new int[`n`];` *Python:* `A = [0] *` n
  Create a new array with n cells, with positions $0, 1, \ldots, n-1$;
  we write $A[0..n) = A[0..n-1]$
- `get(`i`)` *Java/Python*: `A[`i`]`
  Return the content of cell i
- `set(`i`,`x`)` *Java/Python*: `A[`i`] =` x`;`
  Set the content of cell i to x.

⇝ Arrays have *fixed* size (supplied at creation). (≠ `lists` in Python)

Usually directly implemented by compiler + operating system / virtual machine.



Difference to "real" ADTs: ***Implementation usually fixed***
to "a contiguous chunk of memory".

Doubling trick

Can we have unbounded stacks based on arrays? Yes!

Invariants:

- maintain array S of elements, from bottommost to topmost
- maintain index *top* of position of topmost element in `S`
- maintain capacity $C = S.length$ so that $\frac{1}{4}C \leq n \leq C$

$\rightsquigarrow$ can always push more elements!

How to maintain the last invariant?

- before `push`
  If $n = C$, allocate new array of size $2n$, copy all elements.
- after `pop`
  If $n < \frac{1}{4}C$, allocate new array of size $2n$, copy all elements.

$\rightsquigarrow$ ***"Resizing Arrays"***

an implementation technique, not an ADT!

Amortized Analysis

- Any individual operation `push` / `pop` can be expensive!
  $\Theta(n)$ time to copy all elements to new array.
- **But:** One expensive operation of cost T means $\Omega(T)$ next operations are cheap!



distance to boundary

since $n \le C \le 4n$

Formally: consider "credits/potential" $\Phi = \min\{n - \frac{1}{4}C, C - n\} \in [0, 0.6n]$

- amortized cost of an operation = actual cost (array accesses) $- 4 \cdot$ change in Φ
  - cheap `push`/`pop`: actual cost 1 array access, consumes ≤ 1 credits $\rightsquigarrow$ amortized cost ≤ 5
  - copying `push`: actual cost $2n+1$ array accesses, creates $\frac{1}{2}n+1$ credits $\rightsquigarrow$ amortized cost ≤ 5
  - copying `pop`: actual cost $2n+1$ array accesses, creates $\frac{1}{2}n-1$ credits $\rightsquigarrow$ amortized cost 5

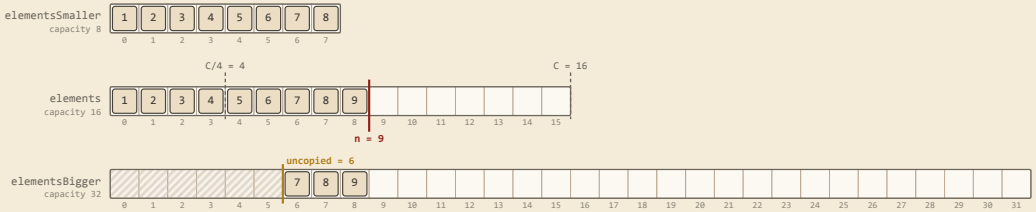
$\rightsquigarrow$ **sequence** of m operations: total actual cost $\le$ total amortized cost + final credits

here: $\le \quad 5m \quad + \quad 4 \cdot 0.6n \quad = \Theta(m+n)$

Deamortized Resizable Arrays

*What if we need $O(1)$ **worst case** time?*

- It's possible to *de-amortize* the resizing arrays solution!
- maintain **3 arrays**: S (as before) and S_2 and $S_{1/2}$ of twice and half the size of S



- write operations go to all 3 arrays
- upon resize, "shift" arrays up/down $\rightsquigarrow$ S_2 resp. $S_{1/2}$ become new S
  - allocate new array, but **delay filling it with elements** ← general strategy!
  - every insert or delete *copies 2 slots* from last resize

$\rightsquigarrow$ by time for next resize, we have caught up and S_2 resp. $S_{1/2}$ ready to use

Deamortized Resizable Arrays – Analysis

Analysis:

assuming memory allocation in $O(1)$ $\rightsquigarrow$ needs to be uninitialized!

- $O(1)$ worst case time for read/write by index, push, and pop!
- up to 7 array accesses per operation
- up to $14n$ space

other time-space trade-offs possible

⚠ we assume here that array allocation is $O(1)$ time (no initialization!)

Rabbit Hole: Can we do this more space-efficiently?

- It might appear as if every efficient implementation of a stack needs $\Omega(n)$ extra space on top of space for storing the n elements in the stack.
- But this is not true!
- Can get operations in $O(1)$ worst-case time with $O(\sqrt{n})$ extra space at any time (!)
  - Maintain a collection of small arrays (plus header with pointers to them)
  - Clever choice of block sizes guarantees
    $O(\sqrt{n})$ blocks of $O(\sqrt{n})$ elements throughout
    and fast calculation of address for an index.

    imaginary "superblocks" of sizes 2^k, $k = 0, 1, \ldots, \lg n$
    kth superblock consists of $2^{k/2}$ actual blocks of $2^{k/2}$ elements each.
  - $O(\sqrt{n})$ extra space is best possible



Brodnik, Carlsson, Demaine, Munro, Sedgewick: *Resizable Arrays in Optimal Time and Space,* WADS **1999**

3.3 Priority Queues & Binary Heaps

Priority Queue ADT – min-oriented version

Now: elements in the bag have different *priorities*.

(Max-oriented) Priority Queue (MaxPQ):

- `construct(`A`)`
  Construct from from elements in array A.
- `insert(`x`,`p`)`
  Insert item x with priority p into PQ.
- `max()`
  Return item with largest priority. (Does not modify the PQ.)
- `delMax()`
  Remove the item with largest priority and return it.
- `changeKey(`x`,`p'`)`
  Update x's priority to p'.
  Sometimes restricted to *increasing* priority.
- `isEmpty()`

Fundamental building block in many applications.



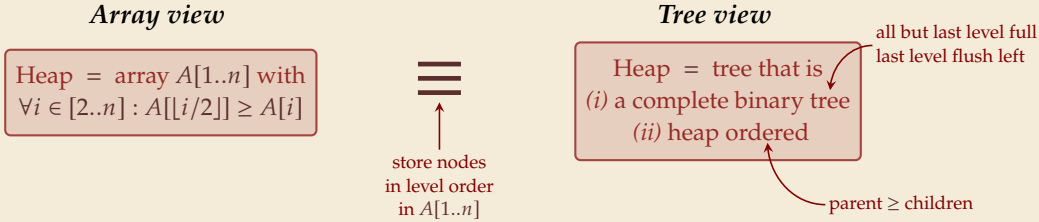
PQ implementations

Elementary implementations

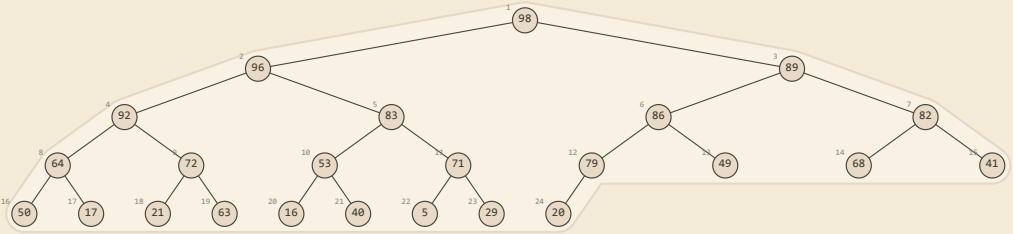
- unordered list $\rightsquigarrow$ $\Theta(1)$ `insert`, but $\Theta(n)$ `delMax`
- sorted list $\rightsquigarrow$ $\Theta(1)$ `delMax`, but $\Theta(n)$ `insert`

Can we get something between these extremes? Like a "slightly sorted" list?

Yes! *Binary heaps.*



Binary heap example





Why heap-shaped trees?

Why complete binary tree shape?

- only one possible tree shape $\rightsquigarrow$ keep it simple!
- complete binary trees have minimal height among all binary trees
- simple formulas for moving from a node to parent or children:
  For a node at index k in A ⚠ Recall: nodes at indices $[1..n]$
  - parent at $\lfloor k/2 \rfloor$ (for $k \geq 2$)
  - left child at $2k$
  - right child at $2k + 1$

Why heap ordered?

- Maximum must be at root! $\rightsquigarrow$ `max()` is trivial!
- But: Sorted only along paths of the tree; leaves lots of leeway for fast inserts

how? . . . stay tuned

3.4 Operations on Binary Heaps

Insert

1. Add new element at only possible place: bottom-most level, next free spot.
2. Let element *swim* up to repair heap order.

Delete Max

1. Remove max (must be in root).
2. Move last element (bottom-most, rightmost) into root.
3. Let root key *sink* in heap to repair heap order.

Heap construction

- n times insert $\rightsquigarrow$ $\Theta(n \log n)$ 
- instead:
  1. Start with singleton heaps (one element)
  2. Repeatedly merge two heaps of height k with new element into heap of height $k + 1$

Analysis

Height of binary heaps:

- *height* of a tree: # edges on longest root-to-leaf path
- *depth/level* of a node: # edges from root $\rightsquigarrow$ root has depth 0
- How many nodes on first k *full* levels? $\sum_{\ell=0}^{k} 2^\ell = 2^{k+1} - 1$

$\rightsquigarrow$ Height of binary heap: $h \;=\; \min k \text{ s.t. } 2^{k+1} - 1 \geq n \;=\; \lfloor \lg(n) \rfloor$

Analysis:

- `insert`: new element "swims" up $\rightsquigarrow$ $\leq h$ steps (h cmps)
- `delMax`: last element "sinks" down $\rightsquigarrow$ $\leq h$ steps ($2h$ cmps)
- `construct` from n elements:

  cost = cost of letting *each node* in heap sink!

$$\leq 1 \cdot h + 2 \cdot (h-1) + 4 \cdot (h-2) + \cdots + 2^\ell \cdot (h-\ell) + \cdots + 2^{h-1} \cdot 1 + 2^h \cdot 0$$

$$= \sum_{\ell=0}^{h} 2^\ell (h-\ell) \;=\; \sum_{i=0}^{h} \frac{2^h}{2^i} i \;=\; 2^h \sum_{i=0}^{h} \frac{i}{2^i} \;\leq\; 2 \cdot 2^h \;\leq\; 2n \text{ steps } (\leq 4n \text{ comparisons})$$

Binary heap summary

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(n)$
`max()`	$O(1)$
`insert(`x`,`p`)`	$O(\log n)$
`delMax()`	$O(\log n)$
`changeKey(`x`,`p'`)`	$O(\log n)$
`isEmpty()`	$O(1)$
`size()`	$O(1)$

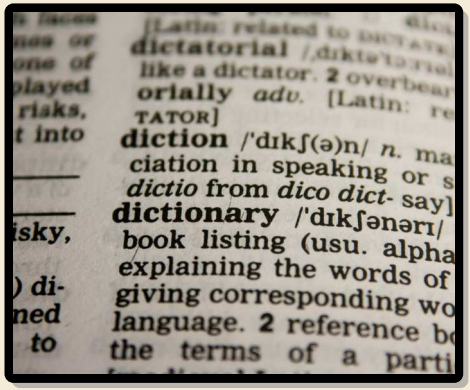
3.5 Symbol Tables

Symbol table ADT

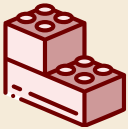
Java: `java.util.Map<K,V>`

Symbol table / Dictionary / Map / Associative array / key-value store:

Python dict `{k:v}`



- `put(`k`,`v`)` *Python* dict: `d[`k`] = `v
  Put key-value pair (k, v) into table
- `get(`k`)` *Python* dict: `d[`k`]`
  Return value associated with key k
- `delete(`k`)` *Python* dict: `del d[`k`]`
  Remove key k (any associated value) form table
- `contains(`k`)` *Python* dict: k `in d`
  Returns whether the table has a value for key k
- `isEmpty(),size()`
- `create()`



Most fundamental building block in computer science.

(Every programming library has a symbol table implementation.)

Symbol tables vs. mathematical functions

- similar interface
- but: mathematical functions are *static/immutable* (never change their mapping)
  (Different mapping is a *different* function)
- symbol table = *dynamic* mapping
  Function may change over time

Elementary implementations

Unordered (linked) list:

- 👍 Fast `put`
- 👎 $\Theta(n)$ time for `get`
- $\rightsquigarrow$ Too slow to be useful

Sorted *linked* list:

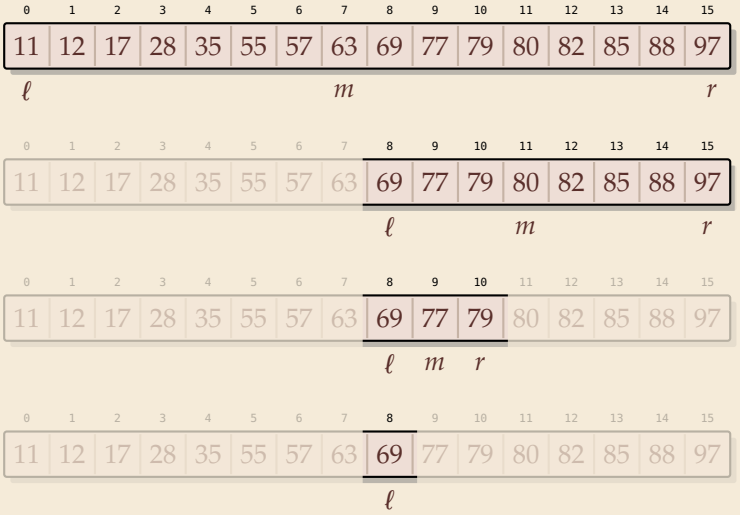
- 👎 $\Theta(n)$ time for `put`
- 👎 $\Theta(n)$ time for `get`
- $\rightsquigarrow$ Too slow to be useful

- $\rightsquigarrow$ *Sorted order does not help us at all?!*

Binary search

*It does help . . . if we have a sorted **array**!*

Example: search for 69



Binary search:

- halve remaining list in each step (± 1)

$\rightsquigarrow \leq \lfloor \lg n \rfloor + 1$ cmps in the worst case



needs random access!

3.6 Binary Search Trees

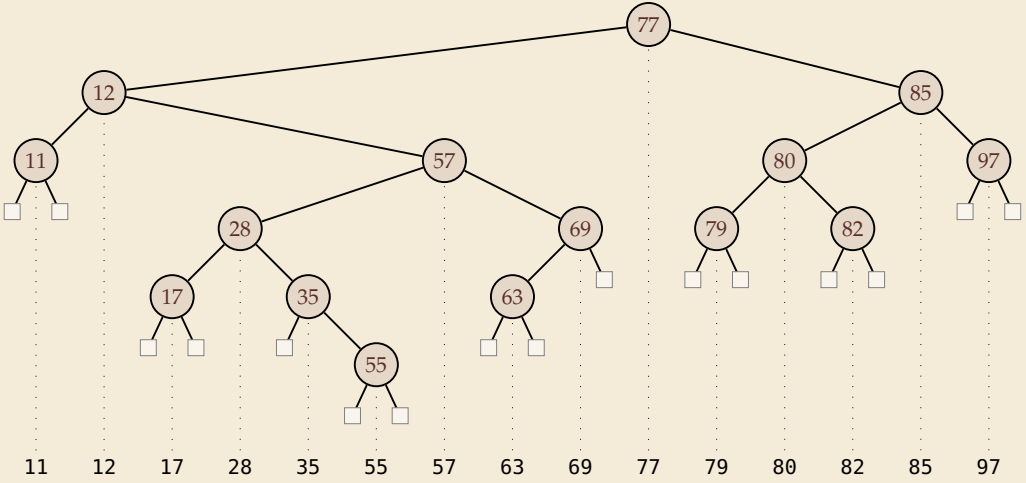
Binary search trees

Binary search trees (BSTs) $\approx$ dynamic sorted array

- binary tree
  - Each node has left and right child
  - Either can be empty (`null`)
- Keys satisfy ***search-tree property***

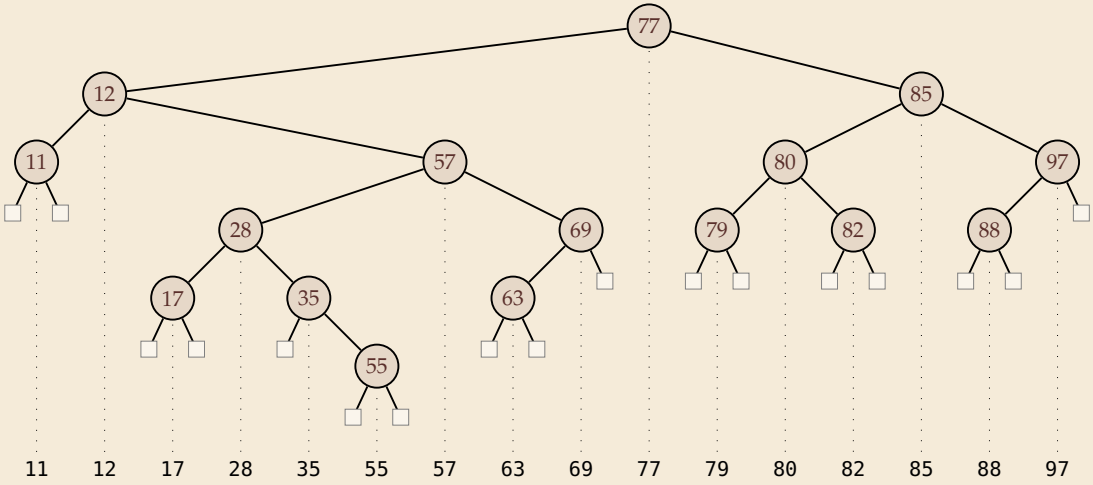
  all keys in left subtree $\leq$ root key $\leq$ all keys in right subtree

BST example & find



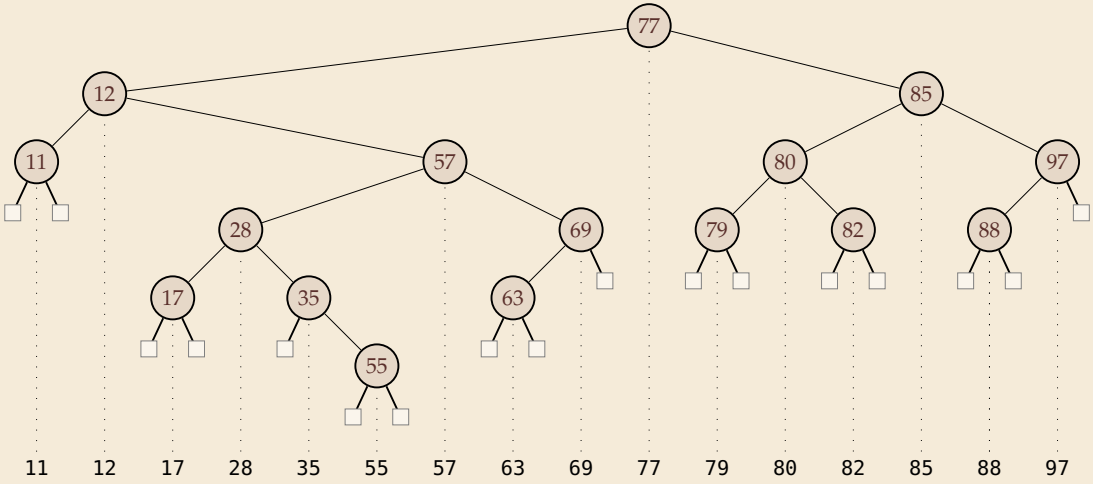
BST insert

Example: Insert 88



BST delete

- Easy case: remove leaf, e. g., 11 ⇝ replace by `null`
- Medium case: remove unary, e. g., 69 ⇝ replace by unique child
- Hard case: remove binary, e. g., 85 ⇝ swap with successor, recurse



Analysis

Let h be the *height* of the BST (longest root-to-leaf path, in edges).

- **Search:** follows one root-to-node path, one comparison per node

$\rightsquigarrow$ $\leq h + 1$ comparisons $\rightsquigarrow$ $O(h)$

- **Insert:** unsuccessful search for the key ($\leq h + 1$ comparisons), then attach a new leaf
  $\rightsquigarrow$ $O(h)$

- **Delete:** search for the key ($O(h)$); leaf / unary case: $O(1)$ pointer changes;
  binary case: find predecessor (walk down the left subtree, $O(h)$), swap, delete the predecessor (leaf or unary) $\rightsquigarrow$ $O(h)$

$\rightsquigarrow$ everything is $O(h)$ . . . *so how large is* h*?*

Random BSTs – average search cost

Model: n distinct keys inserted in *random order* (all $n!$ orders equally likely), no deletions.

Let C_n := expected total cost of searching *each* key once (cost = #nodes on search path).

- root = first inserted key $\rightsquigarrow$ its rank r is uniform in $[1..n]$
- left/right subtree: random BSTs of $r-1$ resp. $n-r$ keys
  (remaining keys still arrive in random order)

$$\rightsquigarrow\ C_n \;=\; n + \frac{1}{n}\sum_{r=1}^{n}\big(C_{r-1}+C_{n-r}\big) \;=\; n + \frac{2}{n}\sum_{k=0}^{n-1} C_k, \qquad C_0 = 0$$

Clever trick: $nC_n - (n-1)C_{n-1} \;=\; 2n - 1 + 2C_{n-1}$ $\rightsquigarrow$ $\dfrac{C_n}{n+1} \;=\; \dfrac{C_{n-1}}{n} + \dfrac{2n-1}{n(n+1)}$ $\rightsquigarrow$

$$\frac{C_n}{n+1} \;=\; \sum_{k=1}^{n}\left(\frac{3}{k+1} - \frac{1}{k}\right) \;=\; 2H_n - 3 + \frac{3}{n+1}$$

$\rightsquigarrow$ $C_n = 2(n+1)H_n - 3n$ $\rightsquigarrow$ average cost per search $\dfrac{C_n}{n} \;\sim\; 2\ln n \;\approx\; 1.39\lg n$

($H_n = \sum_{k\le n}\frac{1}{k} = \ln n + \gamma + O(\frac{1}{n})$)

Not the same as the *height*: $\mathbb{E}[h] \sim 4.311\ln n$ (Devroye 1986) – also $\Theta(\log n)$, but a much harder proof.

BST summary

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(nh)$
`put(`k`,`v`)`	$O(h)$
`get(`k`)`	$O(h)$
`delete(`k`)`	$O(h)$
`contains(`k`)`	$O(h)$
`isEmpty()`	$O(1)$
`size()`	$O(1)$

What is the height of a BST?

Worst Case:

- $h = n - 1 = \Theta(n)$

Average Case:

- Assumption: insertions come in random order
  no deletions

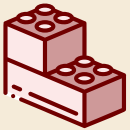
$\rightsquigarrow$ $h = \Theta(\log n)$ in expectation

even "with high probability":
$\forall d\, \exists c : \Pr[h \geq c \lg(n)] \leq n^{-d}$

3.7 Ordered Symbol Tables

Ordered symbol tables

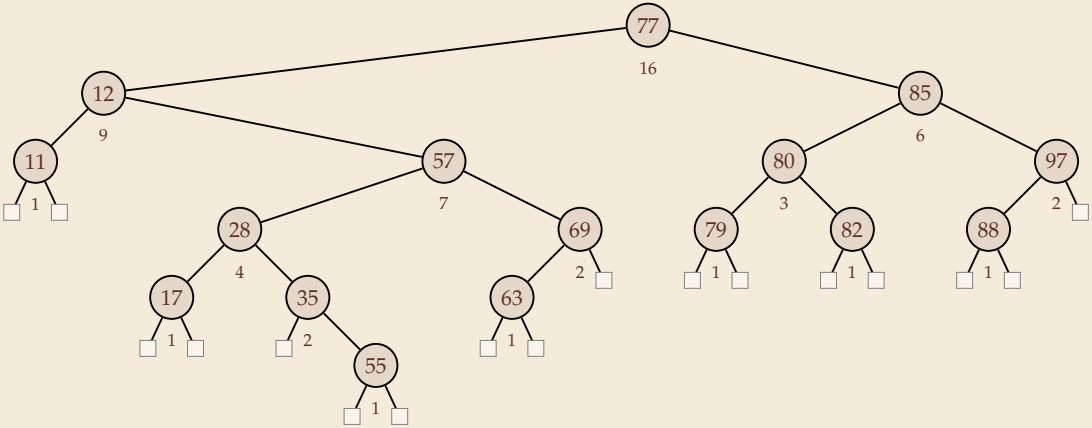
- `min()`, `max()`
  Return the smallest resp. largest key in the ST
- `floor(`x`)`, $\lfloor x \rfloor = \mathbb{Z}$.`floor(`$x$`)`
  Return largest key k in ST with $k \leq x$.
- `ceiling(`x`)`
  Return smallest key k in ST with $k \geq x$.
- `rank(`x`)`
  Return the number of keys k in ST $k < x$.
- `select(`i`)`
  Return the ith smallest key in ST (zero-based, i. e., $i \in [0..n)$)



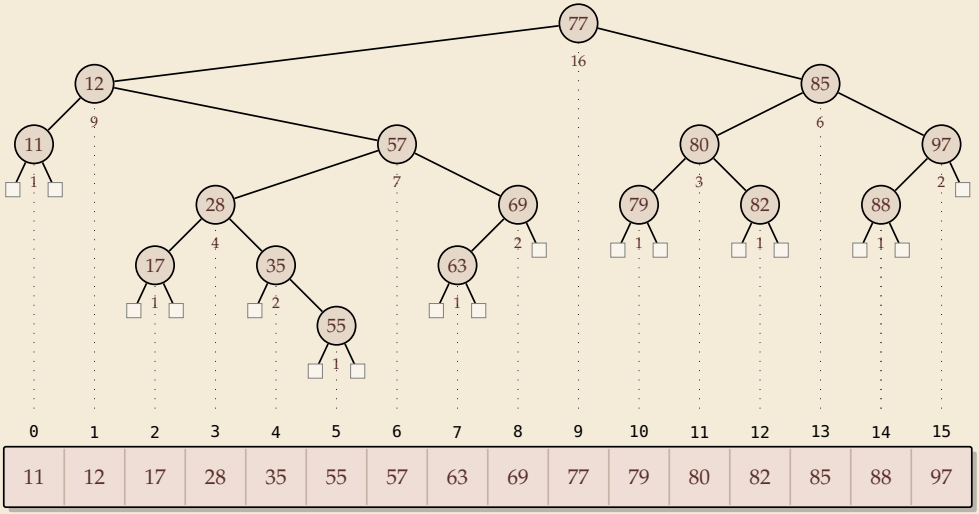
With select, we can simulate access as in a truly dynamic array!

(Might not need any keys at all then!)

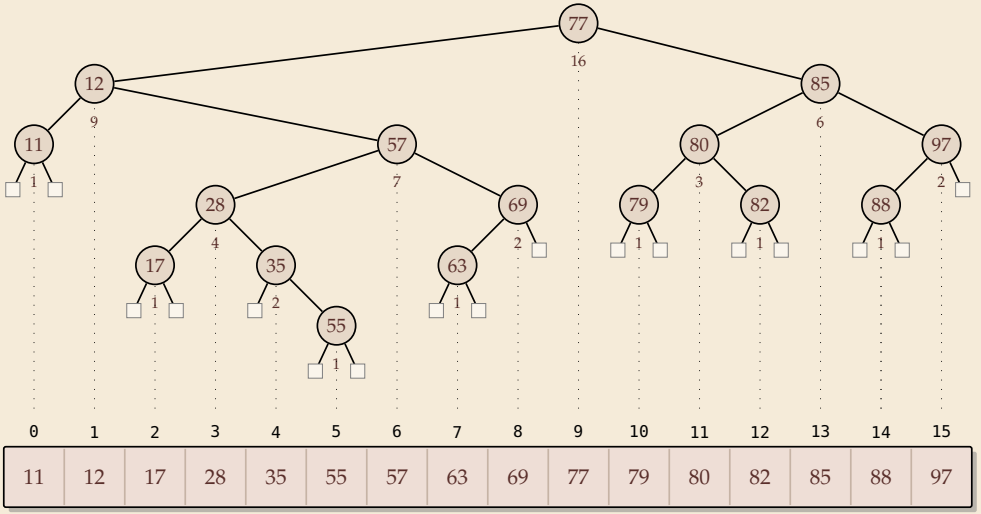
Augmented BSTs



Rank


77
16
12
9
85
6
11
1
57
7
80
3
97
2
28
4
69
2
79
1
82
1
88
1
17
1
35
2
63
1
55
1
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
11 12 17 28 35 55 57 63 69 77 79 80 82 85 88 97

Select


77
16
12
9
85
6
11
1
57
7
80
3
97
2
28
4
69
2
79
1
82
1
88
1
17
1
35
2
63
1
55
1
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
11 12 17 28 35 55 57 63 69 77 79 80 82 85 88 97

Why store subtree sizes?

- Note that in an augmented BST, each node stores the **size of its subtree**.
- . . . why not directly store the **rank**? Would make rank/select much simpler!

- Problem: Single insertion/deletion can change *all* node ranks!

⇝ Cannot efficiently maintain node ranks.



Subtree sizes only change along search path ⇝ $O(h)$ nodes affected

3.8 Balanced BSTs

Balanced BSTs

Balanced binary search trees:

- imposes shape invariant that guarantees $O(\log n)$ height
- adds rules to restore invariant after updates
- many examples known
  - *AVL trees* (height-balanced trees)
  - *red-black trees*
  - *weight-balanced trees* (BB[α] trees)
  - ...



Other options:

I'd love to talk more about all of these ...
(Maybe another time)

- **amortization:** *splay trees*, *scapegoat trees*
  COLA (cache oblivious lookahead array)
- **randomization:** *randomized BSTs*, *treaps*, *skip lists*

BSTs vs. Heaps

Balanced binary search tree

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(n \log n)$
`put(`k`,`v`)`	$O(\log n)$
`get(`k`)`	$O(\log n)$
`delete(`k`)`	$O(\log n)$
`contains(`k`)`	$O(\log n)$
`isEmpty()`	$O(1)$
`size()`	$O(1)$
`min()` / `max()`	$O(\log n) \rightsquigarrow O(1)$
`floor(`x`)`	$O(\log n)$
`ceiling(`x`)`	$O(\log n)$
`rank(`x`)`	$O(\log n)$
`select(`i`)`	$O(\log n)$

~~**Binary heaps**~~ *Strict Fibonacci heaps*

Operation	Running Time
`construct(`$A[1..n]$`)`	$O(n)$
`insert(`x`,`p`)`	~~$O(\log n)$~~ $O(1)$
`delMax()`	$O(\log n)$
`changeKey(`x`,`p'`)`	~~$O(\log n)$~~ $O(1)$
`max()`	$O(1)$
`isEmpty()`	$O(1)$
`size()`	$O(1)$

- apart from faster `construct`, BSTs always as good as binary heaps
- MaxPQ abstraction still helpful
- and faster heaps exist!

3.9 Hashing – Basics

Lower bound for search

The fastest implementations of the ordered symbol table ADT require $\Theta(\log n)$ time to search among n items. *Is this the best possible?*

- **Theorem:** In the comparison model (on the keys), $\Omega(\log n)$ comparisons are required to search a size-n dictionary.

- *Proof:* Any algorithm defines a *decision tree*: internal nodes are comparisons, leaves are outcomes.
  There are $n + 1$ different outcomes ("is x_i" or "not found"), so $\geq n + 1$ leaves;
  a binary tree with ℓ leaves has height $\geq \lceil \lg \ell \rceil$ (ADS). ■

$\rightsquigarrow$ Balanced BSTs are optimal . . . *in the comparison model.*

$\rightsquigarrow$ To beat $\Theta(\log n)$ we must leave that model:
compute with the key instead of only comparing it.

(and we will only get *expected* $O(1)$ – the lower bound is for the worst case)

Hashing Recap

Notation: $U = [0..u)$ universe of keys; $x_1, \ldots, x_n$ the n stored keys;
$T[0..m)$ the table; $\alpha = n/m$ the *load factor*;
$h : U \to [0..m)$

- store x (and its value) at or near $T[h(x)]$

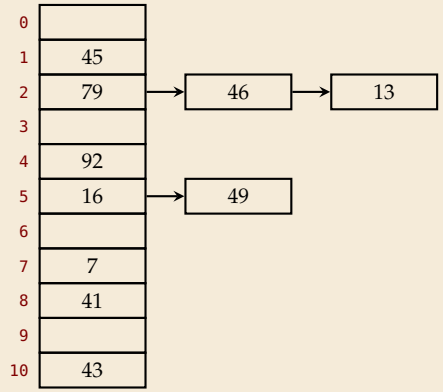
⇝ `get/put/delete` become array accesses . . .

↯ . . . if no two stored keys share an index.

$u \gg m$ ⇝ collisions *unavoidable* (pigeonhole),
and *likely*: already $n \approx 1.18\sqrt{m}$ random keys
collide with probability $\geq \frac{1}{2}$ (birthday paradox)

- Collision handling (details: ADS / S&W §3.4)
  - *separate chaining*: $T[i]$ holds list keys with $h(x) = i$
  - *open addressing*: probe $T[h(x)], T[h(x)+1], \ldots$
    linear, or other probe sequences
    (quadratic, Robin Hood, cuckoo, . . .)

$m = 11$, $h(x) = x \bmod 11$; chaining



Hashing Recap [2]

Notation: $U = [0..u)$ universe of keys; $x_1, \ldots, x_n$ the n stored keys;
$T[0..m)$ the table; $\alpha = n/m$ the *load factor*;
$h : U \to [0..m)$

- *rehash* into a new table when α leaves $[\alpha_{\min}, \alpha_{\max}]$
  $\rightsquigarrow$ $\Theta(n + m)$, amortized $O(1)$ (doubling trick!)

- **Cost** of an operation $\approx$ #keys examined $=$ length of the chain / probe sequence.

  How long are they?

  - in worst case: all keys(!)

Uniform hashing assumption

- ***Simple Uniform Hashing Assumption:***
  For the keys $x_1, \ldots, x_n$ in our dictionary we have

$$\mathop{\mathbb{P}}_{x_i \neq x_j} \left[h(x_i) = h(x_j) \right] \;=\; \frac{1}{m}$$

$\approx$ keys in the input are drawn i.i.d. uniformly from $U = [0..u)$,
and h spreads U evenly over $[0..m)$.

- This is an assumption about the *input*. We will come back to that.

Analysis of chaining (average case)

- L_i: # keys in bucket i
  - how many keys hash to i?
- unsuccessful search: $1 + L_{h(x)}$ memory accesses
- successful search: $\leq 1 + L_{h(x_i)}$ memory accesses

$\rightsquigarrow$ key quantity: $\mathbb{E}[L_{h(x_i)}]$

- Define $X_{i,j} = \mathbb{1}\{h(x_i) = h(x_j)\}$ $\rightsquigarrow$ $L_{h(x_i)} = \sum_{j=1}^{n} X_{i,j}$
  - indicator random variable "keys x_i and x_j collide"
  - deterministically $X_{i,i} = 1$
  - *uniform hashing* implies

    for $i \neq j$: $\mathbb{P}\big[h(x_i) = h(x_j)\big] = \frac{1}{m}$ $(*)$

expectation over the *input*

$$
\begin{aligned}
\mathbb{E}[L_{h(x_i)}] &= \mathbb{E}\Big[\sum_{j=1}^{n} X_{i,j}\Big] \\
&\underset{\text{lin. } \mathbb{E}}{=} \sum_{j=1}^{n} \mathbb{E}[X_{i,j}] \\
&= \sum_{j=1}^{n} \mathbb{P}[X_{i,j} = 1] \\
&= 1 + \sum_{j \neq i} \mathbb{P}[h(x_i) = h(x_j)] \\
&\underset{(*)}{=} 1 + \sum_{j \neq i} \frac{1}{m} \\
&\leq 1 + \alpha
\end{aligned}
$$

3.10 Universal Hashing

Discussion

- Running time for hashing with chaining is $O(1)$ if we keep m proportional to n

$\rightsquigarrow$ see amortized analysis of the *doubling trick*

- `java.util.HashMap` works like this!

  + red-black tree for large buckets

- **But:** only in expectation, and only under the **assumption** of *uniform hashing*!

  - Worst case: all keys collide in one bucket.
  - For *every fixed* h this worst case exists: $u \geq n \cdot m$ $\rightsquigarrow$ some bucket contains $\geq n$ keys of U (pigeonhole)
  - An adversary who knows h inserts exactly those $\rightsquigarrow$ every operation $\Theta(n)$.
    (Not hypothetical: hash-flooding attacks on Perl, PHP, Java `hashCode`, ... – cf. Sedgewick & Wayne)
  - *Can we avoid this?*



Randomization!



Universal families of hash functions

- **Idea:** Choose h **at random** from a set $\mathcal{H}$.
- A set $\mathcal{H} \subseteq \{h : U \to [0..m)\}$ of hash functions is ***c-universal*** if for a (uniformly) random $h \in \mathcal{H}$ we have

$$\forall x \neq x' \; : \; \mathop{\mathbb{P}}_{h \in \mathcal{H}}\big[h(x) = h(x')\big] \; \leq \; \frac{c}{m}$$

- universal = 1-universal
- Compare 1-universal with

Simple Uniform Hashing Assumption:
For the keys $x_1, \ldots, x_n$ in our dictionary we have

$$\mathop{\mathbb{P}}_{x_i \neq x_j}\big[h(x_i) = h(x_j)\big] \; = \; \frac{1}{m}$$

$\rightsquigarrow$ essentially the same collision probability!

But: instead of *hoping* for a benign input (average case),
we *enforce* the average case by randomization – for *every* input.

Analysis of chaining (randomized)

- Choose h *at random* from a c-universal $\mathcal{H}$. *Then* let the adversary pick the keys $x_1, \ldots, x_n$ and x.
- L_i: # keys in bucket i
  - unsuccessful search: $1 + L_{h(x)}$ memory accesses
  - successful search: $\leq 1 + L_{h(x_i)}$ memory accesses
  - $\rightsquigarrow$ key quantity: $\mathbb{E}[L_{h(x_i)}]$
- Define $X_{i,j} = \mathbb{1}\{h(x_i) = h(x_j)\}$ $\rightsquigarrow$ $L_{h(x_i)} = \sum_{j=1}^{n} X_{i,j}$
  - indicator random variable "keys x_i and x_j collide"
  - deterministically $X_{i,i} = 1$
  - *universality* implies

    for $x \neq x'$: $\mathbb{P}_{h \in \mathcal{H}}\big[h(x) = h(x')\big] \leq \frac{c}{m}$ $(*)$

expectation over our *random bits*

$$
\begin{aligned}
\mathbb{E}[L_{h(x_i)}] &= \mathbb{E}\Big[\sum_{j=1}^{n} X_{i,j}\Big] \\
&\underset{\text{lin. } \mathbb{E}}{=} \sum_{j=1}^{n} \mathbb{E}[X_{i,j}] \\
&= \sum_{j=1}^{n} \mathbb{P}[X_{i,j} = 1] \\
&= 1 + \sum_{j \neq i} \mathbb{P}[h(x_i) = h(x_j)] \\
&\underset{(*)}{\leq} 1 + \sum_{j \neq i} \frac{c}{m} \\
&\leq 1 + c\alpha
\end{aligned}
$$

Same analysis – only *what is random* has changed.

Adversary may know $\mathcal{H}$, but not which h we drew; re-draw h at every rehash.

Do universal families exist?

- Yes! e. g., the set of **all** functions $\mathcal{H} = \{h : U \to [0..m)\}$
  - With $h \in \mathcal{H}$ uniform, hash values behave exactly like i. i. d. uniform random variables: $\forall x \in U : h(x) \overset{\mathcal{D}}{=} \text{Uniform}[0..m)$

  $\rightsquigarrow \; \mathop{\mathbb{P}}_{h \in \mathcal{H}}\left[h(x) = h(x')\right] = \frac{1}{m}$

  $\rightsquigarrow$ $\mathcal{H}$ is 1-universal

↯ too expensive! We need $\Theta(u)$ space to store h ...

- Wanted: a *small* family $\rightsquigarrow$ h fits in $O(1)$ words and is $O(1)$ to evaluate

Dot-product hashing

- **Assumption:** m is prime
- **Assumption:** $u = m^r$ for some $r \in \mathbb{N}_{\geq 1}$

$\rightsquigarrow$ key x corresponds to a number with r digits in base m: $x = \langle x_{r-1}, x_{r-2}, \ldots, x_0 \rangle$

- For $a \in [0..u)$, define $h_a(x) = (a^T x) \bmod m = \sum_{l=0}^{r-1} a_l x_l \bmod m$

  dot product of a and x as r-dimensional vectors

- $\mathcal{H} = \{h_a \mid a \in [0..u)\}$

- **Properties:**
  - h_a can be stored in $O(\log u)$ bits (store a!)
  - h_a is computed in $O(1)$ time in the word-RAM with word size $w = \Omega(\log u)$

    $\rightsquigarrow$ keys fit in $O(1)$ words

Dot-product hashing is universal

Theorem 3.1

Let m be prime and $u = m^r$ for $r \in \mathbb{N}_{\geq 1}$. Then $\mathcal{H} = \left\{ h_a \,\middle|\, a \in [0..u) \right\}$ is 1-universal. ◀

Proof:

$x \neq x'$ arbitrary but fixed.

To show: $\mathbb{P}_{a \in [0..m)^r} \left[h_a(x) = h_a(x') \right] \leq \frac{1}{m}$

$x = \langle x_{r-1}, \ldots, x_0 \rangle \neq \langle x'_{r-1}, \ldots, x'_0 \rangle = x'$
$\rightsquigarrow \quad \exists d : x_d \neq x'_d.$

m prime $\rightsquigarrow$ $\mathbb{Z}_m$ is a field
$\rightsquigarrow$ multiplicative inverses:
$\forall y \neq 0 \; \exists y^{-1} : y \cdot y^{-1} \equiv 1 \pmod{m}$ (†)

$$
\begin{aligned}
&\mathbb{P}_{a \in [0..m)^r} \left[h_a(x) = h_a(x') \right] \\
&= \mathbb{P}_{a \in [0..m)^r} \left[\sum_{i=0}^{r-1} a_i x_i \equiv \sum_{i=0}^{r-1} a_i x'_i \pmod{m} \right] \\
&= \mathbb{P}_{a \in [0..m)^r} \left[a_d (x_d - x'_d) \equiv \sum_{i \neq d} a_i (x'_i - x_i) \pmod{m} \right] \\
&\overset{(\dagger)}{=} \mathbb{P}_{a \in [0..m)^r} \left[a_d \equiv \underbrace{(x_d - x'_d)^{-1} \sum_{i \neq d} a_i (x'_i - x_i)}_{C(x, x', a_{\neq d})} \pmod{m} \right] \\
&\overset{\text{law of total prob.}}{=} \mathbb{E}_{a_{\neq d}} \left[\mathbb{P}_{a_d \in [0..m)} \left[a_d \equiv C(x, x', a_{\neq d}) \pmod{m} \,\middle|\, a_{\neq d} \right] \right] \\
&= \mathbb{E}_{a_{\neq d}} \left[\frac{1}{m} \right] = \frac{1}{m} \qquad \blacksquare
\end{aligned}
$$

Shorthand: $a_{\neq d} = \langle a_{r-1}, \ldots, a_{d+1}, a_{d-1}, \ldots, a_0 \rangle$

Linear congruential hashing

Theorem 3.2 (Linear congruential hashing)

Let $p \geq u$ be prime and $m \leq p$. Then

$$\mathcal{H}_{p,m} \;=\; \left\{ h_{a,b} \;\middle|\; a \in [1..p),\ b \in [0..p),\ \ h_{a,b}(x) = \big((ax+b) \bmod p\big) \bmod m \right\}$$

is 1-universal.

Proof:

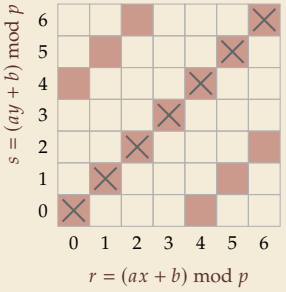
Let $x \neq y$ in U be arbitrary but fixed.

Write $r = (ax+b) \bmod p,\ \ s = (ay+b) \bmod p$.

- We have $r \neq s$: $r - s \equiv a(x-y) \not\equiv 0 \pmod p$, since $a \neq 0$, $x \not\equiv y$ and $p \geq u$ prime.
- Further, $(a,b) \mapsto (r,s)$ is a *bijection* onto the $p(p-1)$ pairs $r \neq s$: $a = (r-s)(x-y)^{-1} \bmod p,\quad b = (r - ax) \bmod p$.

$\rightsquigarrow$ uniform (a,b) = uniform *off-diagonal* cell (r,s).

all pairs (r,s) for $p = 7$, $m = 4$



shaded = collision ($r \equiv s$);
× diagonal = impossible ($r \neq s$).

Proof [2]

Proof (cont.):

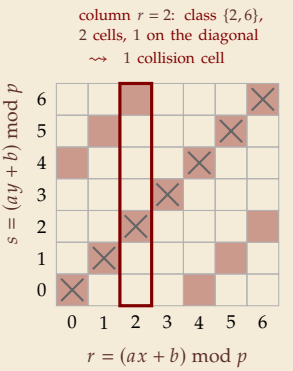
- collision iff $r \equiv s \pmod m$: the shaded cells.

  Per column r: a residue class of $\le \lceil p/m \rceil$ cells, *one of which is the diagonal*

  $\rightsquigarrow \quad \le \lceil p/m \rceil - 1 \le \dfrac{p-1}{m}$ shaded off-diagonal cells.

  shaded off-diagonal: $2+2+2+0 = 6$ of 42 cells $\rightsquigarrow \ \frac{1}{7} \le \frac{1}{4}$ ✓

$\rightsquigarrow \ \mathbb{P}[h_{a,b}(x) = h_{a,b}(y)] \ \le \ \dfrac{p \cdot \frac{p-1}{m}}{p(p-1)} \ = \ \dfrac{1}{m}.$



all pairs (r, s) for $p = 7$, $m = 4$

Note: *Crude count* (all shaded cells, diagonal included, $a = 0$ allowed): 13 of $49 \approx 0.27 > \frac{1}{4}$ $\rightsquigarrow$ only c-universal with $c = \left(\lceil \frac{p}{m} \rceil / \frac{p}{m}\right)^2 \le 4$. The bijection is what lets us drop the diagonal.

Other universal families

Theorem 3.3 (Multiplicative hashing)

For w-bit keys and $m = 2^d$, the family
$\left\{ h_z(x) = \left((z \cdot x) \bmod 2^w\right) \operatorname{div} 2^{w-d} \;\middle|\; z \text{ odd},\ z \in [1..2^w) \right\}$ is 2-universal. ◀



Dietzfelbinger, Hagerup, Katajainen, Penttonen: *A Reliable Randomized Algorithm for the Closest-Pair Problem*, J. Algorithms **1997**

In practice: Python randomizes its string hash per process (SipHash, since 3.3) for exactly this reason; Java does not, and instead turns long chains into red-black trees.

Hashing with chaining – Summary

Average case → expectation in the worst case

- With a *universal hash family* we achieve the same complexity as under the uniform hashing assumption –

  but: without any assumption on the input!

- analogous to Quicksort with random pivot choice (Unit 4)

- total running time is still amortized (doubling trick for rehashing!)


fastest (amortized) access time


simple data structure


somewhat wasteful with space


worst case $O(n)$ per operation (cuckoo hashing fixes this for searches)

Comparison of Classic Hashing Schemes

Hash table design	Search hit	Search miss	Insert	Space	good $\alpha = \frac{n}{m}$
Separate Chaining	$\approx 1 + \frac{1}{2}\alpha$	$\approx \alpha$	= miss	$n + m$	≈ 2
Linear Probing	$\approx \frac{1}{2}\left(1 + \frac{1}{1-\alpha}\right)$	$\approx \frac{1}{2}\left(1 + \frac{1}{(1-\alpha)^2}\right)$	= miss	m	≤ 0.5
Quadratic Probing	$\approx 1 + \ln\left(\frac{1}{1-\alpha}\right) - \frac{1}{2}\alpha$	$\approx \frac{1}{1-\alpha} - \alpha + \ln\left(\frac{1}{1-\alpha}\right)$	= miss	m	≤ 0.7
Robin Hood (Lin. P.)	= Lin. P.; Var. $O(1)$	$\leq$ Lin. P.	= miss	m	≤ 0.9
Robin Hood Clever	$O(1)$	$O(1)$	= miss	m	≤ 1 (=any!)
d-way Cuckoo H.	$\leq d$ worst case	$\leq d$ worst case	amort.	m	$< c_d$

$c_2 = \frac{1}{2}$, $c_3 \approx 0.918$

- Assumption: uniform hashing (all m^n hash sequences equally likely)
- Cost: expected # (equality) comparisons, for $n, m \to \infty$ with α fixed
- Space usage in words on top of space for items (without space for optional optimizations)
- Robin Hood only *reorders* within clusters: same mean as linear probing, but the cost of a search is *concentrated* (and with random probing + "mean-centred" search even $O(1)$ expected in a full table [Poblete & Viola 2019])

More improvements possible with word-RAM bitwise tricks $\rightsquigarrow$ *Advanced Data Structures*

Hashing vs. Balanced Search Trees

Advantages of Balanced Search Trees

- $O(\log n)$ worst-case operation cost
- Does not require any assumptions, special functions, or known properties of input distribution
- Predictable (and often smaller) space usage (exactly n nodes)
- Never need to rebuild the entire structure
- supports ordered dictionary operations (rank, select etc.)

Advantages of Hash Tables

- $O(1)$ operations (if hashes well-spread and load factor small)
- We can choose space-time tradeoff via load factor
- Cuckoo hashing achieves $O(1)$ worst-case for search & delete